

One Period Model

θ is the no.of risky assets, c is consumption, x is initial wealth

$$X^{x,\theta}_1 = (x - c - S^1_0\theta)(1 + r) + S^1_0\theta Z_1$$

Target : maximize the Expectation of Utility of consumption that is financed on Wealth and Investment,

$$\mathbb{E}[U_0(c) + U_1(X_1)] = \mathbb{E}[U_0(c) + U_1((x - c - S^1_0\theta)(1 + r) + S^1_0\theta Z_1)]$$

$$\theta \in I_y \qquad I_y = \left[\frac{y}{S^1_0((1 + r) - a)}, \frac{-y}{S^1_0(b - (1 + r))} \right]$$

If $a = 0$, then $\theta S^1_0 \leq (x - c)$, so the investor cannot borrow money to purchase assets. If he borrows money to purchase assets then he will go into debt when the assets price fall down to 0 and the utility will be $-\infty$. Borrowing here means $\theta S^1_0 > (x - c)$, he has more money to spend on assets after the consumption.

$$y = (x - c)(1 + r) \qquad \tilde{V}_1(y; \theta) = U[y + \theta S^1_0(Z - (1 + r))]$$

Lemma 1 : For $y > 0$, $\tilde{V}_1(y; \theta)$ is concave in θ on I_y

Note : $\tilde{V}_1(0; y) = U_1(y) < \infty$. By concavity of $\tilde{V}_1(\cdot \text{ , } y)$ it cannot take the value $+\infty$

Lemma 2 : $\tilde{V}_1^*(y)$ is concave and increasing in y on $[0, \infty]$ where $\tilde{V}_1^*(y) = \sup_{\theta \in I_y} \tilde{V}_1(y; \theta)$

The problem of maximising the objective function over c and θ is now reduced to maximising

$$U_0(c) + \tilde{V}_1^*((x - c)(1 + r))$$

over $c \in [0, x]$, where U_0 and $\tilde{V}_1^*$ are concave functions. The supremum will be attained, possibly at an endpoint.

Lemma 3 : V is concave in x

$$V(x) = \sup_{c \in [0, x]} \left\{ U_0(c) + \tilde{V}_1^*((x - c)(1 + r)) \right\}$$

Multi-period Model

For $(U_i)_{i=0, \dots, N}$ a family of increasing concave functions $U_i : \mathbb{R}_+ \rightarrow [-\infty, \infty)$, consider the objective function

$$\mathbb{E} \left[\sum_{j=0}^{N-1} U_j(C_j) + U_N(X_N) \right]$$

$$X_k = (X_{k-1} - C_{k-1} - \Theta_{k-1}S^1_{k-1})(1 + r) + \Theta_{k-1}S^1_{k-1}Z_k$$

subject to $X_0 = x$, where $(Z_k)_{k=1, 2, \dots, N}$ is an adapted family of i.i.d. random variables such that Z_k is $\mathcal{F}_k$ -measurable. C represents consumption, and $\#$ the trading strategy expressed as number of shares.

$$\mathbb{E} \left[\mathbb{E} \left[\sum_{j=0}^{N-1} U_j(C_j) + U_N(X_N) \middle| \mathcal{F}_{N-1} \right] \right] = \mathbb{E} \left[\sum_{j=0}^{N-2} U_j(C_j) + \mathbb{E}[U_{N-1}(C_{N-1}) + U_N(X_N) | \mathcal{F}_{N-1}] \right]$$

$$\begin{aligned} & \sup_{(C_j)_{0 \leq j \leq N-1}; (\Theta_j)_{1 \leq j \leq N}} \mathbb{E} \left[\sum_{j=0}^{N-1} U_j(C_j) + U_N(X_N) \right] \\ &= \sup_{(C_j)_{0 \leq j \leq N-2}; (\Theta_j)_{1 \leq j \leq N-1}} \mathbb{E} \left[\sum_{j=0}^{N-2} U_j(C_j) + \sup_{C_{N-1}, \Theta_N} \mathbb{E}[U_{N-1}(C_{N-1}) + U_N(X_N) | \mathcal{F}_{N-1}] \right] \end{aligned}$$

But, the problem of finding $\sup_{C_{N-1}, \Theta_N} \mathbb{E}[U_{N-1}(C_{N-1}) + U_N(X_N) | \mathcal{F}_{N-1}]$ over the last

time-period is exactly like the one-period case. Moreover, the solution $\tilde{U}_{N-1} = \tilde{\tilde{U}}_{N-1}(X_{N-1})$ is a concave function. Hence, by solving the problem over the final N^{th} period we can reduce the optimization problem to

$$\mathbb{E} \left[\sum_{j=0}^{N-2} U_j(C_j) + \tilde{U}_{N-1}(X_{N-1}) \right]$$

Dynamic Programming Equation

Utility Functions

CRRA : $U(x) = \frac{x^{1-R}}{1-R}$, $R \in (0, \infty) \setminus \{1\}$; $U'(x) = x^{-R}$; RRA $= R$

Log ($R \rightarrow 1$) : $U(x) = A \ln x + B$

Discrete Wealth

$X_1 = (x - c - \varepsilon S^1_0)(1 + r) + \varepsilon S^1_0Z$

No-arbitrage : $\mathbb{P}(Z < 1 + r) > 0$ & $\mathbb{P}(Z > 1 + r) > 0$

Bellman Equation

$V_k(x) = \inf_u \{K(k, x, u) + \mathbb{E}[V_{k+1}(F(k, x, u, Z))]\}$

HJB Equation (continuous)

$\sup_u \{K + \frac{1}{2}a^2V_{xx} + bV_x + \dot{V}\} = 0$

1-Period CRRA

$c^* = \hat{\alpha}^*x$; $(\hat{\alpha}^*)^{-1} = 1 + \frac{1}{1+r} \left(\frac{A_0}{\kappa A_1(1+r)} \right)^{-1/R}$

$\kappa^{1-R} = \sup_{\lambda} \frac{1}{1-R} \mathbb{E}[(1 + \lambda(Z - (1 + r)))^{1-R}]$

$$V(x) = \Xi \frac{x^{1-R}}{1-R}, \quad \Xi = \left(A_0^{1/R} + (\kappa A_1)^{1/R} (1 + r)^{1/R-1} \right)^R$$

***N*-Period CRRA Recursion**

$$\Xi_N = A_N; \quad \Xi_{k-1} = \left(A_{k-1}^{1/R} + (\kappa \Xi_k)^{1/R} (1 + r)^{1/R-1} \right)^R$$

∞ -Horizon CRRA

$\theta := \kappa^{1/R} \beta^{1/R} (1 + r)^{1/R-1}$; well-posed iff $\theta < 1$

$\Xi = A(1 - \theta)^{-R}$; $\hat{\alpha}^* = 1 - \theta$

N-Period Log

$v_k(x) = \alpha_k \ln x + \beta_k$; $\alpha_k = \sum_{j=k}^N A_j$

$C_k^* = \frac{A_k}{\alpha_k} X_k$; ϕ^* constant (time-invariant)

Merton (CRRA, perpetual)

$\hat{\pi} = \frac{\rho}{\sigma \bar{R}}$; $\eta = \frac{1}{\bar{R}} \left(\delta - r(1 - R) - \frac{\rho^2(1-R)}{2\bar{R}} \right)$

$\hat{C}_t = \eta X_t$; $\hat{V}(x) = \eta^{-R} \frac{x^{1-R}}{1-R}$; w.p. iff $\eta > 0$

Merton (log, perpetual)

$A = 1/\delta$; $\hat{C}_t = \delta X_t$; $\hat{\pi} = \frac{\mu - r}{\sigma^2 \delta}$

Dual Approach

$\zeta_t = e^{-\rho W_t - (r + \rho^2/2)t}$; budget : $\mathbb{E}[\int_0^\infty \zeta_t C_t \, \mathrm{d}t] \leq x$

FOC : $e^{-\delta t} U'(\hat{C}_t) = \gamma \zeta_t$; $\hat{C}_t = I(\gamma e^{\delta t} \zeta_t)$; $V_x = \gamma$

Whittle’s Equation

$G'' - (G')^2 + 1 = 0$; sub $g = e^{-G}$ gives $g'' - g = 0$

Optimal : $U_t^* = -G'(X_t)$

[What is the Merton Problem?] The **Merton problem** asks : given initial wealth x invested in a financial market (riskless bond + risky asset), how should an agent optimally split wealth between *consumption* and *investment* over time to maximise expected utility? Named after Merton (1969, 1971).

Base case elements :

- Finite time horizon with bequest, or infinite horizon.
- Discrete or continuous time.
- Constant-parameter, complete financial model (one or d risky assets).
- Simple parametric utility : typically CRRA or logarithmic.

Extensions : stochastic volatility/interest rates, transaction costs, short-sale constraints, habit formation.

Two fundamental questions :

- When is the problem *well-posed* (finite value)?
- How* should the agent invest and consume (optimal strategy)?

The Discrete-Time Case

One-Period Model

Financial market :

- Riskless bond : $S^0_0 \xrightarrow{1} S^0_1 = S^0_0(1 + r)$.

- Risky asset : $S^1_0 \xrightarrow{p(\omega_i)} S^1_1(\omega_i) = S^1_0 Z(\omega_i)$, where $Z \geq 0$, $\mathbb{E}[Z] > 1 + r$.
- No-arbitrage condition** : $\mathbb{P}(Z < 1 + r) > 0$ *and* $\mathbb{P}(Z > 1 + r) > 0$.

Agent’s choices :

- $c \geq 0$: consumption at time 0.
- ε : number of shares of S^1 held over $(0, 1)$.

[Wealth Equation - One Period]

$$X_1 = (x - c - \varepsilon S^1_0)(1 + r) + \varepsilon S^1_0 Z = (x - c)(1 + r) + \varepsilon S^1_0 (Z - (1 + r)).$$

Objective : Maximise $\mathbb{E}[U_0(c) + U_1(X_1)]$ over (c, ε) with U_i increasing, concave.

Admissibility : With $a \leq Z \leq b$, $0 \leq a < 1 + r < b \leq \infty$:

$$\frac{-(x - c)(1 + r)}{S^1_0(b - (1 + r))} \leq \varepsilon \leq \frac{(x - c)(1 + r)}{S^1_0((1 + r) - a)}.$$

If $b = \infty$ then $\varepsilon \geq 0$; if $a = 0$ then $\varepsilon S^1_0 \leq (x - c)$ (no borrowing to buy shares).

Reduction : Define $\tilde{V}_1(\varepsilon; y) := \mathbb{E}[U_1(y + \varepsilon S^1_0(Z - (1 + r)))]$ and $\tilde{V}_1^*(y) := \sup_{\varepsilon \in \mathcal{I}_y} \tilde{V}_1(\varepsilon; y)$.

[Key Lemmas] **Lemma 1** : $\tilde{V}_1(\varepsilon; y)$ is *concave* in ε on $\mathcal{I}_y$ (follows from concavity of U_1).

Lemma 2 : $\tilde{V}_1^*(y)$ is *concave and increasing* in $y \geq 0$.

Lemma 3 : $V(x) = \sup_{c \in [0, x]} \{U_0(c) + \tilde{V}_1^*((x - c)(1 + r))\}$ is *concave* in x .

Proofs : All use concavity of U_1 + Jensen’s inequality, plus ε -optimal strategies for ε arguments.

[Value Function and Interior Optimum]

$$V(x) = \sup_{c \in [0, x]} \left\{ U_0(c) + \tilde{V}_1^*((x - c)(1 + r)) \right\}.$$

Interior optimum condition (assuming smoothness) :

$$U'_0(c^*) = (1 + r)(\tilde{V}_1^*)'((x - c^*)(1 + r)).$$

Multi-Period Model

Work on filtered space $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_k)_{0 \leq k \leq N})$. (Z_k) are i.i.d. $\mathcal{F}_k$ -measurable.

Wealth equation :

$$X_k = (X_{k-1} - C_{k-1} - \varepsilon_{k-1}S^1_{k-1})(1 + r) + \varepsilon_{k-1}S^1_{k-1}Z_k, \quad X_0 = x.$$

Objective : Maximise $\mathbb{E} \left[\sum_{j=0}^{N-1} U_j(C_j) + U_N(X_N) \right]$ over adapted (C, ε) with $X \geq 0$.

[Principle of Backward Induction] The N -period problem reduces to a sequence of one-period problems :

- Solve the final period $[N - 1, N]$: obtain $\tilde{U}_{N-1}(X_{N-1})$ (concave in X_{N-1}).
- This replaces U_N in the $(N - 1)$ -period problem. Repeat inductively.
- At each step the resulting value function is concave, ensuring well-posedness of the next step.

Key : At each stage we solve $\sup_{C_{k-1}, \varepsilon_{k-1}} \mathbb{E}[U_{k-1}(C_{k-1}) + V_k(X_k) \mid \mathcal{F}_{k-1}]$, a one-period problem.

The Dynamic Programming / Bellman Equation

[Controlled Markov Process Setup] $X_{n+1} = F(n, X_n, U_n, Z_{n+1})$ with (Z_k) i.i.d. independent of past controls.

Goal : minimise $\mathbb{E} \left[\sum_{n=0}^{N-1} K(n, X_n, U_n) + \xi(X_N) \right]$ over admissible controls $U \in \mathcal{U}$.

Define $V_N(x) = \xi(x)$ and $V_k(x) = \inf_{(U_j)_{j \geq k}} \mathbb{E} \left[\sum_{n=k}^{N-1} K(n, X_n, U_n) + \xi(X_N) \mid X_k = x \right]$.

[Bellman / Dynamic Programming Equation]

$$V_k(x) = \inf_u \left\{ K(k, x, u) + \mathbb{E}[V_{k+1}(F(k, x, u, Z))] \right\}.$$

Solved backwards from terminal condition $V_N(x) = \xi(x)$. Optimal control $u^*(k, x)$ achieves the infimum.

Key properties :

- The infimum over u at each step depends only on x (Markovian optimal control).
- For maximisation problems (e.g. Merton), replace inf with sup and K by $-K$.
- Concavity of V_k is preserved backwards if K and F are suitably structured.

CRRA Utility - Discrete Time

[CRRA Definition and Properties] **Power law utility** : $U(x) = \frac{x^{1-R}}{1-R}$, $R \in (0, \infty) \setminus \{1\}$.

$U'(x) = x^{-R} > 0$; $U''(x) = -R x^{-R-1} < 0$ (increasing, strictly concave).

Absolute risk aversion : $-U''(x)/U'(x) = R/x$.

Relative risk aversion (RRA) : $-xU''(x)/U'(x) = R$ (constant - hence CRRA).

Log utility ($R = 1$) : $U'(x) = 1/x$ gives $U(x) = A \ln x + B$.

One-Period CRRA

Problem : $\sup_{c, \varepsilon} \left\{ \frac{A_0}{1-R} c^{1-R} + \frac{A_1}{1-R} \mathbb{E} \left[((x - c - \varepsilon S^1_0)(1 + r) + \varepsilon S^1_0 Z)^{1-R} \right] \right\}$.

Normalisation : Write $\lambda = \varepsilon S^1_0/y$ (fraction of post-consumption wealth placed in risky asset, normalised). Define :

$$\kappa^{1-R} := \sup_{\lambda \in \Lambda} \frac{1}{1-R} \mathbb{E}[(1 + \lambda(Z - (1 + r)))^{1-R}],$$

where $\Lambda = \{-1/(b - (1 + r)), 1/((1 + r) - a)\}$ (the admissibility interval for λ , independent of y).

[One-Period CRRA Value Function]

$$\tilde{V}_1^*(y) = \frac{\kappa A_1}{1-R} y^{1-R}, \quad y = (x - c)(1 + r).$$

$$V(x) = \sup_c \left\{ \frac{A_0}{1-R} c^{1-R} + \frac{\kappa A_1(1 + r)^{1-R}}{1-R} (x - c)^{1-R} \right\}.$$

Optimal consumption : $A_0 c^{-R} = \kappa A_1(1 + r)(x - c)^{-R}(1 + r)^{1-R}$:

$$\hat{c}^* = \hat{\alpha}^*x, \qquad (\hat{\alpha}^*)^{-1} = 1 + \frac{1}{1 + r} \left(\frac{A_0}{\kappa A_1(1 + r)} \right)^{-1/R}.$$

Optimal investment : $\varepsilon^* S^1_0 = \lambda^*(x - c^*)(1 + r)$ where λ^* does not depend on wealth x .

Key insight : It is optimal to invest a **constant fraction of post-consumption wealth** in the risky asset.

[One-Period Value Function (CRRA)]

$$V(x) = \frac{\Xi x^{1-R}}{1-R}, \qquad \Xi = \left(A_0^{1/R} + (\kappa A_1)^{1/R} (1 + r)^{1/R-1} \right)^R.$$

The value function has the same CRRA form with the same risk aversion R .

N-Period CRRA

$U_j(c) = \frac{A_j}{1-R} c^{1-R}$, $j = 0, \dots, N$; (Z_k) i.i.d.

Backward induction ansatz : $V_k(x) = \Xi_k \frac{x^{1-R}}{1-R}$, true for $k = N$ with $\Xi_N = A_N$.

[N-Period CRRA Backward Recursion] Given $V_k(x) = \Xi_k \frac{x^{1-R}}{1-R}$, we obtain $V_{k-1}(x) = \Xi_{k-1} \frac{x^{1-R}}{1-R}$ with :

$$\Xi_{k-1} = \left(A_{k-1}^{1/R} + (\kappa \Xi_k)^{1/R} (1 + r)^{1/R-1} \right)^R \tag{8}$$

$$\hat{\alpha}_{k-1}^* = \left(1 + (1 + r)^{1/R-1} \left(\frac{A_{k-1}}{\kappa \Xi_k} \right)^{-1/R} \right)^{-1} \tag{7}$$

Optimal investment fraction λ^* does **not depend on k** (same at all time steps).

Infinite Horizon CRRA

Objective : $\sup_{C, \varepsilon} \mathbb{E} \left[\sum_{k=0}^\infty A \beta^k \frac{C_k^{1-R}}{1-R} \right]$ with $X \geq 0$, $X_0 = x$.

Ansatz : $V_k(x) = \Xi \beta^k \frac{x^{1-R}}{1-R}$ (consistent with stationarity).

Plugging into recursion (8) with $\Xi_k = \Xi \beta^k$ and $\Xi_{k-1} = \Xi \beta^{k-1}$:

[Infinite Horizon Well-Posedness and Solution] Define $\theta := \kappa^{1/R} \beta^{1/R} (1 + r)^{1/R-1}$.

From $\Xi^{1/R} \beta^{(k-1)/R} = A^{1/R} \beta^{(k-1)/R} + \kappa^{1/R} \Xi^{1/R} \beta^{k/R} (1 + r)^{1/R-1}$:

$$\Xi^{1/R} = A^{1/R} (1 - \theta)^{-1} \implies \boxed{\Xi = A(1 - \theta)^{-R}}. \tag{9}$$

Well-posedness condition : $\theta < 1$, i.e.

$$\kappa^{1/R} \beta^{1/R} (1 + r)^{1/R-1} < 1 \iff \beta < \frac{1}{\kappa(1 + r)^{1-R}}. \tag{10}$$

Optimal consumption fraction : $\hat{\alpha}^* = 1 - \theta > 0$ (requires well-posedness).

Optimal investment fraction : same λ^* as the one-period problem, independent of time.

[Ill-Posedness - Remark 1] If (10) fails and $R < 1$: $\exists$ admissible strategy with $J = +\infty$ (supremum is $+\infty$).

If (10) fails and $R > 1$: every admissible strategy gives $J = -\infty$ (supremum is $-\infty$).

Verification - Infinite Horizon Discrete

Define the process $M = (M_k)_{k \geq 0}$ by :

$$M_k = \sum_{j=0}^{k-1} \frac{A\beta^j C_j^{1-R}}{1-R} + \beta^k V(X_k), \quad V(y) = \Xi \frac{y^{1-R}}{1-R}.$$

[Verification Theorem - Infinite Horizon Discrete] Suppose :

- $M^{x,\varepsilon,C}$ is a **supermartingale** for every admissible (ε,C) .
- $M^{x,\hat{\varepsilon},\hat{C}}$ is a **martingale** for the candidate optimal strategy.
- Transversality** : $\liminf_{k \rightarrow \infty} \beta^k \mathbb{E}[V(X_k^{x,\varepsilon,C})] \geq 0$ for all admissible strategies.
- $\lim_{k \rightarrow \infty} \beta^k \mathbb{E}[V(X_k^{x,\hat{\varepsilon},\hat{C}})] = 0$ for the candidate optimal strategy.

Then V is the value function and $(\hat{\varepsilon},\hat{C})$ is optimal.

Note : Condition (3) is automatic if $R < 1$ (utility non-negative).

Proof sketch :

From (1) : $V(x) = M_0 \geq \mathbb{E}[M_k] \geq \mathbb{E}\left[\sum_{j=0}^{k-1} \frac{A\beta^j C_j^{1-R}}{1-R}\right] + \beta^k \mathbb{E}[V(X_k)]$.

By Monotone Convergence + (3) : $V(x) \geq \mathbb{E}\left[\sum_{j=0}^{\infty} \frac{A\beta^j C_j^{1-R}}{1-R}\right] = J(x,C)$.

From (2) : equality holds for $(\hat{\varepsilon},\hat{C})$ + (4) : $V(x) = J(x,\hat{C})$. Hence V = value function.

Log Utility - Discrete Time (Q3-Q6)

One-Period Log Utility (Q3)

With $U_0(c) = A_0 \ln c + B_0$, $U_1(x) = A_1 \ln x + B_1$, use the fraction parametrisation $\varepsilon S_0^1 = \pi(x - c)$:

$$X_1 = (x - c)(\pi Z + (1 - \pi)(1 + r)).$$

The objective separates :

$$\max_{c,\pi}\{A_0 \ln c + B_0 + A_1 \ln(x - c) + A_1 \mathbb{E}[\ln(\pi Z + (1 - \pi)(1 + r))] + B_1\}.$$

[Log Utility One-Period Solution] **Optimal consumption** : $\frac{A_0}{c^*} - \frac{A_1}{x - c^*} = 0 \implies c^* = \frac{A_0}{A_0 + A_1}x$.

Optimal portfolio fraction : $\pi^* = \arg \max_{\pi} \mathbb{E}[\ln(\pi Z + (1 - \pi)(1 + r))]$.

Optimal shares : $\varepsilon^* S_0^1 = \pi^*(x - c^*) = \frac{\pi^* A_1}{A_0 + A_1}x$.

Value function : $V(x) = (A_0 + A_1) \ln x + D$ for some constant D .

Envelope Theorem (Q2 & Q4)

[Envelope Result : $V'(x) = U'_0(c^*(x))$] Differentiating $V(x) = \sup_{c,\varepsilon} \{U_0(c) + \mathbb{E}[U_1(X_1)]\}$ using the envelope theorem and the first-order condition $U'_0(c^*) = (1 + r)\mathbb{E}[U'_1(X_1^*)]$:

$$V'(x) = \mathbb{E}[U'_1(X_1^*)](1 + r) = U'_0(c^*(x)).$$

For log : $V'(x) = A_0/c^*(x) = (A_0 + A_1)/x$, confirming $V(x) = (A_0 + A_1) \ln x + D$.

N-Period Log Utility (Q5)

Objective : $\mathbb{E}\left[\sum_{k=0}^{N-1} (A_k \ln C_k + B_k) + A_N \ln X_N + B_N\right]$.

Ansatz : $v_k(x) = \alpha_k \ln x + \beta_k$, with $\alpha_N = A_N$, $\beta_N = B_N$.

[N-Period Log Bellman Recursion] **Bellman equation** :

$$v_k(x) := \sup_{0 \leq c \leq x, \phi \in I_{k+1}} \{A_k \ln c + B_k + \mathbb{E}[v_{k+1}((x - c)((1 + r) + \phi(Z_{k+1} - (1 + r)))]\}.$$

Separates into : $\sup_c \{A_k \ln c + \beta_{k+1} \ln(x - c)\}$ and $\sup_{\phi} \mathbb{E}[\ln((1 + r) + \phi(Z - (1 + r)))]$.

Optimal consumption : $c_k^*(x) = \frac{A_k}{A_k + \alpha_{k+1}}x$, giving $\alpha_k = A_k + \alpha_{k+1} = \sum_{j=k}^N A_j$.

$$C_k^* = \frac{A_k}{\sum_{j=k}^N A_j} X_k.$$

Optimal portfolio : $\phi_k^* = \phi^* = \arg \max_{\phi} \mathbb{E}[\ln((1 + r) + \phi(Z - (1 + r)))]$ (same for all k).

Special case (Q5) : If $A_j = A\rho^j$ ($j < N$), $A_N = A\rho^N\Phi$, then $C_k^*/X_k = (1 + \rho + \dots + \rho^{N-k}\Phi)^{-1}$.

For this to be independent of k : $\Phi = 1 + \rho\Phi$, i.e. $\Phi = \frac{1}{1-\rho}$. Then $C_k^*/X_k = 1 - \rho$.

Infinite Horizon Log Utility (Q6)

With $A_j = A\rho^j$ and $\Phi = 1/(1 - \rho)$, the infinite horizon problem gives constant proportional consumption $\hat{C}_k = (1 - \rho)X_k$. The candidate optimal policy for infinite horizon is obtained as the limit of the N -period policy as $N \rightarrow \infty$.

Continuous Time - Setup

Financial Model

- Riskless bond : $dS_t^0/S_t^0 = r \, dt$ (rate r).
- Risky asset : $dS_t^1/S_t^1 = \sigma \, dW_t + \mu \, dt$, W Brownian motion, $\sigma > 0$.
- Sharpe ratio : $\varrho = (\mu - r)/\sigma$.

Wealth Equation and Admissibility

Agent controls : consumption rate $C_t \geq 0$ and portfolio fraction π_t (fraction of wealth in risky asset).

[Continuous-Time Wealth SDE]

$$\frac{dX_t}{X_t} = \pi_t \sigma \, dW_t + [r + \pi_t(\mu - r)] \, dt - \frac{C_t}{X_t} \, dt.$$

In expanded form :

$$dX_t = X_t \pi_t \sigma \, dW_t + (X_t[r + \pi_t(\mu - r)] - C_t) \, dt, \quad X_0 = x.$$

A strategy (π,C) is **admissible** if $X_t^{x,\pi,C} \geq 0$ for all t . Let $\tau = \inf\{u : X_u = 0\}$; then $\pi_t = C_t = 0$ for $t \geq \tau$ and $X_t = 0$ for $t \geq \tau$.

Objective Function

Finite horizon with bequest : $\mathbb{E}\left[\int_0^T U(s,C_s) \, ds + B(X_T)\right]$.

Perpetual (infinite horizon) : $\mathbb{E}\left[\int_0^{\infty} U(s,C_s) \, ds\right]$.

CRRa utility (used throughout) : $U(s,c) = e^{-\delta s} \frac{c^{1-R}}{1-R}$, $R \in (0,\infty) \setminus \{1\}$.

The Martingale Optimality Principle

[General Setup] Controlled diffusion : $dX_t = a(t,X_t,U_t) \, dW_t + b(t,X_t,U_t) \, dt$, $X_0 = x$.

Return function (one-signed) : $J(x,U) = \mathbb{E}\left[\int_0^{\infty} K(s,X_s,U_s) \, ds\right]$.

Define : $M_t^{U,V} = \int_0^t K(s,X_s,U_s) \, ds + V(t,X_t^{x,U})$.

[Theorem 1 - Martingale Optimality Principle] Suppose $V \in \mathcal{C}^{1,2}$ and control U^* such that :

- $M^{U,V}$ is a **supermartingale** for all admissible U .
- $M^{U^*,V}$ is a **martingale**.
- $\liminf_{t \rightarrow \infty} \mathbb{E}[V(t,X_t^{x,U})] \geq 0$ for all admissible U .
- $\lim_{t \rightarrow \infty} \mathbb{E}[V(t,X_t^{x,U^*})] = 0$.

Then $\sup_U J(x,U) = V(0,x)$ and U^* is optimal.

Proof (Q9 version) :

- Supermartingale : $V(0,x) = M_0^U \geq \mathbb{E}[M_t^U]$, so $V(0,x) \geq \liminf_t \mathbb{E}[\int_0^t K \, ds] + \liminf_t \mathbb{E}[V(t,X_t)]$.
- By Monotone Convergence + (iii) : $V(0,x) \geq J(x,U)$ for all U .
- Martingale + (iv) : $V(0,x) = \lim_t \mathbb{E}[M_t^{U^*}] = \mathbb{E}[\int_0^{\infty} K(\cdot,U^*) \, ds] = J(x,U^*)$.
- Equality throughout : $V(0,x) = \inf_U J(x,U) \leq J(x,U^*) = V(0,x)$.

The HJB Equation

By Itô's formula, $dM_t^{U,V} = K \, dt + aV_x \, dW_t + (\mathcal{L}V) \, dt$ where :

$$\mathcal{L}v(t,x,u) = \frac{a(t,x,u)^2}{2} v_{xx} + b(t,x,u)v_x + \dot{v}.$$

[Hamilton-Jacobi-Bellman (HJB) Equation] For M^U supermartingale in general and martingale for U^* :

$$\sup_u \left\{ K(t,x,u) + \frac{1}{2} a(t,x,u)^2 V_{xx} + b(t,x,u)V_x + \dot{V} \right\} = 0.$$

Optimal control : $U^*(t,x) = \arg \max_u \{ \mathcal{L}V + K \}$.

Note : The HJB gives only a *candidate* solution. Conditions (i)-(iv) of Theorem 1 must still be verified.

Whittle's Flypaper Example

[Problem Setup] Process : $X_t^U = x + B_t + \int_0^t U_s \, ds$, $0 < x < z$, B = Brownian motion.

Let $\tau = \tau_{0,z}^{X^U} = \inf\{s : X_s = 0 \text{ or } X_s = z\}$. **Minimise** :

$$E\left[\int_0^{\tau} \frac{1}{2} (1 + U_s^2) \, ds + k(X_{\tau}^U)\right]$$

where $k(0)$ and $k(z)$ are given positive constants ($k > 0$).

Approach : Let $M_t^U = \int_0^{t \wedge \tau} \frac{1 + U_s^2}{2} \, ds + G(X_{t \wedge \tau}^U)$ for function G to be determined.

Require $G(0) = k(0)$, $G(z) = k(z)$, and (by Itô) :

$$\inf_u \left\{ \frac{1}{2} G''(x) + uG'(x) + \frac{1}{2} (1 + u^2) \right\} = 0.$$

Optimal control : Differentiate in u : $G'(x) + u = 0$, so $\hat{U}_t = -G'(X_t)$.

Substituting : $G'' - (G')^2 + 1 = 0$.

[Solution of Whittle's ODE] **Substitution** : $g(x) = e^{-G(x)}$. Then $g'' = ((G')^2 - G'')g$ and using (18) :

$$g'' - g = 0.$$

General solution : $g(x) = A_0 \cosh x + A_1 \sinh x$.

Boundary conditions : $g(0) = e^{-k(0)}$, $g(z) = e^{-k(z)}$:

$$A_0 = e^{-k(0)}, \quad A_1 = e^{-k(z)} \operatorname{cosech} z - e^{-k(0)} \coth z.$$

$$G(x) = -\ln(e^{-k(0)} \cosh x + [e^{-k(z)} \operatorname{cosech} z - e^{-k(0)} \coth z] \sinh x).$$

(Since $g > 0$ on $[0,z]$, G is well-defined and twice differentiable.)

Verification : $M_t^* = \int_0^{t \wedge \tau^*} \frac{1 + (U_s^*)^2}{2} \, ds + G(X_{t \wedge \tau^*}^{U^*})$, $dM_t^* = G'(X_t^{U^*}) \, dB_t \cdot \mathbf{1}_{t \leq \tau^*}$.

Since G' is bounded : $[M^*]_t \leq \tau^* \|G'\|_{\infty}^2$. Using $\mathbb{E}[\tau^*] < \infty$ (from $\frac{1}{2} \mathbb{E}[t \wedge \tau^*] \leq M_0^* = G(x)$), M^* is L^2 -bounded, hence a UI martingale, and :

$$G(x) = M_0^* = \mathbb{E}[M_{\tau^*}^*] = \mathbb{E}\left[\int_0^{\tau^*} \frac{1 + (U_s^*)^2}{2} \, ds + k(X_{\tau^*}^{U^*})\right].$$

For a general control, M^U is a submartingale (i.e. $\mathbb{E}[M_t^U] \geq G(x)$), so $G(x) \leq$ any admissible cost. Hence G is the value function and $\hat{U}_t = -G'(X_t)$ is optimal.

The Perpetual Merton Problem - Continuous Time

Problem Formulation

Find : $\sup_{(\pi,C) \in \mathcal{A}(x)} J(C)$ where $J(C) = \mathbb{E}\left[\int_0^{\infty} e^{-\delta s} \frac{C_s^{1-R}}{1-R} \, ds\right]$ and wealth solves (19).

Candidate via HJB

Scaling / homogeneity argument : If $(\pi,C) \in \mathcal{A}(x)$, then $(\pi,\varsigma C) \in \mathcal{A}(\varsigma x)$ for $\varsigma > 0$, giving

$H(\varsigma x) = \varsigma^{1-R} H(x)$. Therefore $H(x) = A \frac{x^{1-R}}{1-R}$ and :

$$V(t,x) = Ae^{-\delta t} \frac{x^{1-R}}{1-R}. \quad (20)$$

HJB equation (with $\varrho = (\mu - r)/\sigma$) :

$$\sup_{\pi,c} \left\{ e^{-\delta t} \frac{c^{1-R}}{1-R} + \frac{1}{2} \pi^2 \sigma^2 x^2 V_{xx} + [x(r + \varrho \sigma \pi) - c] V_x + \dot{V} \right\} = 0. \quad (21)$$

Using (20), dividing by $e^{-\delta t}$ and taking suprema :

- Over c : $c^{-R} = Ax^{-R}$ so $\hat{c} = A^{-1/R}x$.
- Over π : $\pi \sigma^2 x^2 V_{xx} + \varrho \sigma x V_x = 0$ so $\hat{\pi} = \frac{\varrho}{\sigma R}$.

Substituting back, the HJB = 0 becomes :

[Merton Constant η and Optimal Solution] Define :

$$\eta := \frac{1}{R} \left(\delta - r(1 - R) - \frac{\varrho^2(1 - R)}{2R} \right). \quad (23)$$

Provided $\eta > 0$ (well-posedness condition) :

$$A = \eta^{-R}; \quad \hat{\pi} = \frac{\varrho}{\sigma R}; \quad \hat{C}_t = \eta X_t; \quad \hat{V}(x) = \eta^{-R} \frac{x^{1-R}}{1 - R}.$$

Constant Proportional Strategies

Strategy $(\pi_t = \pi, C_t = \varsigma X_t)$: wealth is :

$$X_t = x \exp \left(\pi \sigma W_t + \left(r + \pi(\mu - r) - \varsigma - \frac{\pi^2 \sigma^2}{2} \right) t \right). \quad (24)$$

[F -Function and Objective under Proportional Strategy]

$$F(\pi,\varsigma) := \delta - (1 - R) \left(r + \varrho \sigma \pi - \frac{\pi^2 \sigma^2}{2R} - \varsigma \right). \quad (27)$$

If $F(\pi,\varsigma) > 0$:

$$J(\varsigma X) = \frac{x^{1-R}}{1 - R} \cdot \frac{\varsigma^{1-R}}{F(\pi,\varsigma)}. \quad (28)$$

Check : $F(\hat{\pi},\eta) = \eta > 0$, so $J(\hat{C}) = \hat{V}(x)$. ✓

Sensitivity of η to Parameters (Q7)

$$\frac{\partial \eta}{\partial \delta} = \frac{1}{R} > 0; \quad \frac{\partial \eta}{\partial r} = -\frac{1 - R}{R} \text{ (either sign);} \quad \frac{\partial \eta}{\partial (\varrho^2)} = -\frac{1 - R}{2R^2} \text{ (either sign).}$$

[Interpretation]

- More impatient ($\delta \uparrow$) $\Rightarrow$ consume more ($\eta \uparrow$). Always.
- Higher r : if $R < 1$ (risk-tolerant), invest more $\Rightarrow$ consume more; if $R > 1$, opposite.
- Higher Sharpe ratio : sign depends on R vs. 1.
- Graphs : if $\delta > r$, η is positive at $\pi = 0$ and decreasing; if $\delta < r + \varrho^2/2$, a turning point exists.

Fiat Verification

Definition 0.1 (Fiat Admissibility). $(\pi,C) \in \mathcal{A}(x)$ is **flat admissible** ($\mathcal{A}^F(x)$) if :

(P) $X^{x,\pi,C} > 0$ $\mathbb{P}$ -a.s.

(M) The local martingale $\int_0^{\cdot} e^{-\delta t} \sigma \pi_t (X_t^{x,\pi,C})^{1-R} \, dW_t$ is a supermartingale.

(T) Transversality : $\liminf_{t \rightarrow \infty} \mathbb{E}\left[e^{-\delta t} \frac{(X_t^{x,\pi,C})^{1-R}}{1-R}\right] \geq 0$.

[Theorem 2 - Fiat Optimality] Suppose $\eta > 0$. Then $V^F(x) := \sup_{C \in \mathcal{C}^F(x)} J(C) = J(\hat{C}) = \hat{V}(x)$,

with $\hat{\pi} = \varrho/(\sigma R)$ and $\hat{C}_t = \eta X_t^{x,\hat{\pi},\hat{C}}$.

Proof outline :

- Set $M_t = \int_0^t e^{-\delta s} \frac{C_s^{1-R}}{1-R} \, ds + e^{-\delta t} \hat{V}(X_t)$. Apply Itô.
- $dM_t = dN_t + e^{-\delta t} \mathcal{L}(\pi_t,C_t;X_t,\hat{V}) \, dt$ where N is a local martingale.
- Compute $\mathcal{L}(\hat{\pi},\hat{c};x,\hat{V}) = 0$ (HJB satisfied) and $\mathcal{L}(\pi,c;x,\hat{V}) \leq 0$ for all (π,c) .
- So $M_t \geq \hat{V}(x) + N_t$. Fiat condition (M) : N is a supermartingale $\Rightarrow \mathbb{E}[M_t] \geq \hat{V}(x)$.
- Taking $t \rightarrow \infty$ with Monotone Convergence + (T) : $J(C) \leq \hat{V}(x) = J(\hat{C})$.

Log Utility - Continuous Time (Q8)

Problem : Maximise $\mathbb{E}\left[\int_0^{\infty} e^{-\delta s} \ln C_s \, ds\right]$.

Ansatz : $V(t,x) = Ae^{-\delta t} \ln x + Be^{-\delta t}$.

By Itô on $M_t = \int_0^t e^{-\delta s} \ln C_s \, ds + V(t,X_t)$:

$$\begin{aligned} dM_t &= e^{-\delta t} \left[\ln C_t - \delta (A \ln X_t + B) + A \frac{dX_t}{X_t} - \frac{A}{2} \frac{d[X]_t}{X_t^2} \right] \\ &= e^{-\delta t} \left[(\ln \varsigma - A\varsigma) - \delta B + Ar + A\varrho \sigma \pi + Ar - \frac{A}{2} \sigma^2 \pi^2 \right] dt + (\text{martingale}) \end{aligned}$$

where $\varsigma_t = C_t/X_t$.

[Log Utility Continuous-Time Solution] Require $\delta A = 1 \Rightarrow A = 1/\delta$.

Optimal consumption rate : $\sup_{\varsigma} (\ln \varsigma - A\varsigma) = 0$ at $\varsigma = A^{-1} = \delta$:

$$\hat{C}_t = \delta X_t.$$

Optimal portfolio : $A\varrho\sigma - A\sigma^2\hat{\pi} = 0$:

$$\hat{\pi} = \frac{\varrho}{\sigma\delta} = \frac{\mu - r}{\sigma^2\delta}.$$

Constant B : from $\ln \delta - 1 - \delta B + r/\delta + \varrho^2/(2\delta) = 0$:

$$B = \frac{1}{\delta} \left(\frac{r}{\delta} + \frac{\varrho^2}{2\delta} - 1 + \ln \delta \right).$$

Verification needs : $\mathbb{E}[e^{-\delta t} \ln X_t^*] \rightarrow 0$; M^* is a martingale; for general strategies, $\limsup \mathbb{E}[e^{-\delta t} \ln X_t] \leq 0$ (so M is a supermartingale).

The Dual Approach

State-Price Density

[State-Price Density]

$$\zeta_t = e^{-rt} \mathcal{E}(-\varrho W)_t = \exp\left(-\varrho W_t - \left(r + \frac{\varrho^2}{2}\right)t\right). \quad (35)$$

Equivalently, $d\zeta_t = -\varrho\zeta_t \, dW_t - r\zeta_t \, dt$ (i.e. $\zeta_t = Z_t/S_t^0$ where $dZ_t = -\varrho Z_t \, dW_t$).

[Proposition 1 - Budget Feasibility] A non-negative progressively measurable $C \in \mathcal{C}(x)$ must satisfy :

$$\mathbb{E}\left[\int_0^\infty \zeta_s C_s \, ds\right] \leq x. \quad (36)$$

(Sufficiency holds by Brownian martingale representation theorem.)

Proof of necessity : Define $Y_t = \zeta_t X_t + \int_0^t \zeta_s C_s \, ds$.

By product rule and SDEs for ζ and X :

$$dY_t = \zeta_t X_t (\pi_t \sigma - \varrho) \, dW_t.$$

Y is a non-negative local martingale, hence a supermartingale. So :

$$x = Y_0 \geq \liminf_{t \rightarrow \infty} \mathbb{E}[Y_t] \geq \mathbb{E}\left[\int_0^\infty \zeta_s C_s \, ds\right].$$

Dual Verification Steps

[Dual Approach - Complete Algorithm]

- FOC** : $e^{-\delta t} U'(\hat{C}_t) = \gamma \zeta_t$ for multiplier $\gamma > 0$.
- Invert** : $\hat{C}_t = I(\gamma e^{\delta t} \zeta_t)$ where $I = (U')^{-1}$.
- Budget equality** : Choose γ such that $\mathbb{E}\left[\int_0^\infty \zeta_s \hat{C}_s \, ds\right] = x$.
- Optimality** : By concavity tangent inequality $U(b) \leq U(a) + (b-a)U'(a)$:

$$\mathbb{E} \int e^{-\delta t} U(C_t) \, dt \leq \mathbb{E} \int e^{-\delta t} U(\hat{C}_t) \, dt + \underbrace{\gamma \mathbb{E} \int \zeta_t (C_t - \hat{C}_t) \, dt}_{\leq 0 \text{ by budget}}$$

Hence $J(C) \leq J(\hat{C})$ for all admissible C .

- Identify** : $V_x(x) = \gamma(x)$; integrate to recover $V(x)$.

CRRA via Dual

$U(x) = \frac{x^{1-R}}{1-R}$, $I(y) = (U')^{-1}(y) = y^{-1/R}$.

$\hat{C}_t = (\gamma e^{\delta t} \zeta_t)^{-1/R} = \gamma^{-1/R} e^{-\delta t/R} \zeta_t^{-1/R}$.

$\mathbb{E}[\hat{C}_t \zeta_t] = \gamma^{-1/R} e^{-\eta t}$ where $\eta = \frac{1}{R}(\delta - r(1-R) - \frac{\varrho^2(1-R)}{2R})$.

Budget : $\gamma^{-1/R}/\eta = x \Rightarrow \gamma^{-1/R} = \eta x \Rightarrow \hat{C}_0 = \eta x$.

$V_x(x) = \gamma(x) = \eta^{-R} x^{-R} \Rightarrow V(x) = \eta^{-R} \frac{x^{1-R}}{1-R}$. ✓

Duality Question - Non-Standard Utility

[Duality Question : $U(x) = 2 \ln x + 2\sqrt{1+x} - 2 \ln(1 + \sqrt{1+x})$] **Setting** : Perpetual Merton with rate r , Sharpe ratio λ , discount rate $\delta > \frac{r+\lambda^2}{2}$. Define $\psi := 2\delta - r - \lambda^2 > 0$.

(a) U increasing and concave :

$U'(x) = \frac{2}{x} + \frac{1}{\sqrt{1+x}} - \frac{1}{\sqrt{1+x}(1+\sqrt{1+x})} = \frac{2}{x} + \frac{1}{1+\sqrt{1+x}} = \frac{1+\sqrt{1+x}}{x} > 0$.

Set $u = \sqrt{1+x} - 1 > 0$, so $x = u^2 + 2u$ and $U'(x) = 1/u > 0$. Concavity : $U''(x) < 0$. ✓

(b) Inverse of $f(y) = \frac{2a}{y} + \frac{b^2}{y^2}$:

$y = f^{-1}(\omega) = g(\omega) = \frac{a+\sqrt{a^2+b^2\omega}}{\omega}$.

(c) Antiderivative : $\int^x \frac{\sqrt{a^2+b^2\omega}}{\omega} d\omega = a \ln x + 2\sqrt{a^2+b^2x} - 2a \ln(a + \sqrt{a^2+b^2x})$.

(d) $I = (U')^{-1}$: With $U'(x) = 1/u$ and $x = u^2 + 2u$: $I(y) = (1/y)^2 + 2(1/y) = \frac{1}{y^2} + \frac{2}{y}$.

Optimal consumption : $\hat{C}_t = I(\gamma e^{\delta t} \zeta_t) = \frac{1}{\gamma^2 \zeta_t^2 e^{2\delta t}} + \frac{2}{\gamma \zeta_t e^{\delta t}}$.

Budget constraint computation :

$\zeta_t^{-1} = \exp(\lambda W_t + (r + \lambda^2/2)t)$, so $\mathbb{E}[\zeta_t^{-2}] = e^{(r+\lambda^2)t}$ and $\mathbb{E}[\zeta_t^{-1}] = e^{rt+\lambda^2 t/2} \dots$

Evaluating : $x = \frac{2}{\delta\gamma} + \frac{1}{\psi\gamma^2}$ where $a = 1/\delta$, $b^2 = 1/\psi$.

Solving for γ : $\gamma = g(x) = \frac{1/\delta+\sqrt{(1/\delta)^2+x/\psi}}{x}$.

Value function : $V_x(x) = \gamma(x)$. Integrating (using part (c)) :

$$V(x) = \frac{2}{\delta} \ln x + 2\sqrt{\frac{1}{\delta^2} + \frac{x}{\psi}} - \frac{2}{\delta} \ln\left(\frac{1}{\delta} + \sqrt{\frac{1}{\delta^2} + \frac{x}{\psi}}\right) + D.$$

Dice Game / Optimal Stopping (Problem Sheet Q1)

[Optimal Stopping : Dice Rolling with N Rounds] **Setup** : Roll a fair die up to N times. At each round you may stop (keep current score) or continue.

Final round ($k = N$) : Must take score. Expected value = 7/2.

Round $N-1$: Stop if roll $> 7/2$, i.e. roll ≥ 4 (= 4,5,6). Otherwise continue.

$$\text{Average} = \frac{1}{6}(6+5+4+3 \cdot \frac{7}{2}) = \frac{51}{12} = \frac{17}{4}.$$

Round $N-2$: Stop if roll $\geq 17/4 \approx 4.25$, i.e. roll 5 or 6. Game value = $\frac{1}{6}(4 \cdot \frac{17}{4} + 5 + 6) = \frac{14}{3}$.

Round $N-3$: Stop if roll 5 or 6. Game value = $\frac{1}{6}(4 \cdot \frac{14}{3} + 5 + 6) = \frac{89}{18}$.

Round $N-4$: Stop if roll 5 or 6. Game value = $\frac{1}{6}(4 \cdot \frac{89}{18} + 5 + 6) = \frac{277}{54} \approx 5.13$.

General : In all rounds sufficiently far from the end, stop if and only if roll is 6. As $N \rightarrow \infty$, value $\rightarrow$ some limit > 5 .

Connection to DP : This is the Bellman equation with $V_k(x) = \max(x, \mathbb{E}[V_{k+1}(\text{roll})])$. Each step, compare current score with continuation value.

Summary Tables

Setting	Value Function $V(x)$	Opt. Consumption	Opt. Investment
1-period CRRA	$\Xi \frac{x^{1-R}}{1-R}$	$\hat{\alpha}^* x$	Fraction λ^* of $(x - c^*)(1+r)$
N -period CRRA	$\Xi_k \frac{x^{1-R}}{1-R}$ (recursion)	$\hat{\alpha}_{k-1}^* X_{k-1}$ (varies)	Same λ^* every step
∞ -horiz CRRA	$A(1-\theta)^{-R} \frac{x^{1-R}}{1-R}$	$(1-\theta)X_k$	Same λ^* always
1-period log	$(A_0 + A_1) \ln x + D$	$\frac{A_0}{A_0+A_1} x$	ϕ^* (time-invariant)
N -period log	$\alpha_k \ln x + \beta_k$	$\frac{A_k}{\alpha_k} X_k$	ϕ^* all steps
Merton CRRA (cts)	$\eta^{-R} \frac{x^{1-R}}{1-R}$	ηX_t	$\hat{\pi} = \frac{\varrho}{\sigma R}$
Merton log (cts)	$\frac{1}{\delta} \ln x + B$	δX_t	$\hat{\pi} = \frac{\mu-r}{\sigma^2\delta}$
Whittle	$G(x)$ via ODE	-	$\hat{U}_t = -G'(X_t)$

(37)

Technique	When to use and key steps
Backward induction	Multi-period discrete models. Solve from period N backwards. Value fn. inherits concavity.
CRRA ansatz	Guess $V_k(x) \propto x^{1-R}/(1-R)$. Verify constants via recursion (8).
Log ansatz	Guess $v_k(x) = \alpha_k \ln x + \beta_k$. $\alpha_k = \sum_{j=k}^N A_j$. c and ϕ problems separate.
HJB	Continuous time. Use scaling to guess form of V , compute $\hat{\pi}$ and $\hat{c}$, solve for constants.
Dual approach	Write FOC, invert U' , use budget constraint, identify $\gamma = V_x$. Great for non-CRRA utilities.
Martingale verif.	Show M is supermartingale in general, martingale for $\hat{U}$, verify transversality.

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